

eyemR-Vis: A Mixed Reality System to Visualise Bi-Directional Gaze Behavioural Cues Between Remote Collaborators

Allison Jing
University of South Australia

Kieran May
University of South Australia

Mahnoor Naeem
University of South Australia

Gun Lee
University of South Australia

Mark Billingham
University of South Australia

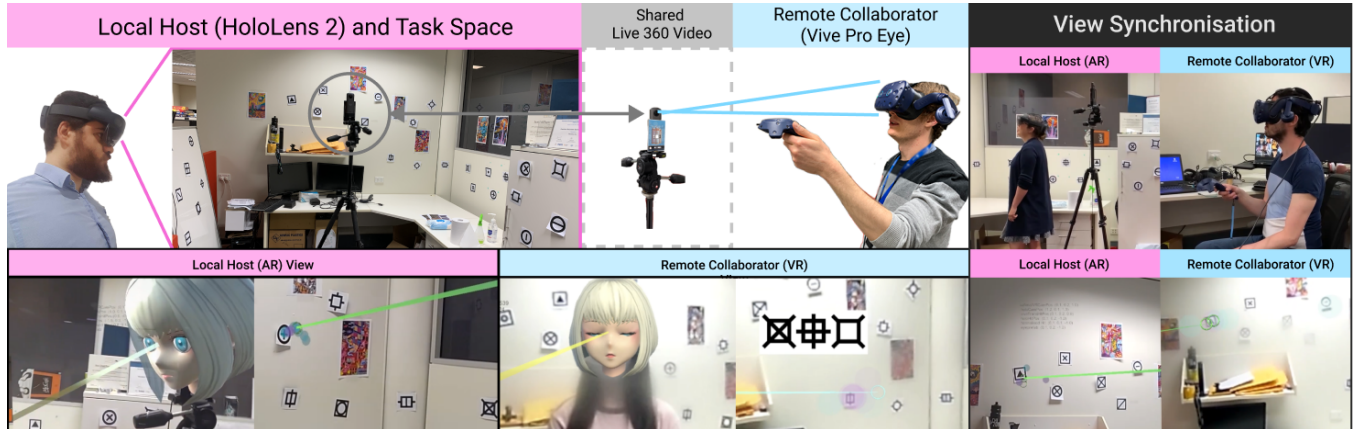


Figure 1: The eyemR-Vis prototype system, showing an AR user (HoloLens2) sharing gaze cues with a VR user (HTC Vive Pro Eye)

ABSTRACT

This demonstration shows eyemR-Vis, a 360 panoramic Mixed Reality collaboration system that translates gaze behavioural cues to bi-directional visualisations between a local host (AR) and a remote collaborator (VR). The system is designed to share dynamic gaze behavioural cues as bi-directional spatial virtual visualisations between a local host and a remote collaborator. This enables richer communication of gaze through four visualisation techniques: browse, focus, mutual-gaze, and fixated circle-map. Additionally, our system supports simple bi-directional avatar interaction as well as panoramic video zoom. This makes interaction in the normally constrained remote task space more flexible and relatively natural. By showing visual communication cues that are physically inaccessible in the remote task space through reallocating and visualising the existing ones, our system aims to provide a more engaging and effective remote collaboration experience.

CCS CONCEPTS

• **Human-centered computing** → **Visualization design and evaluation methods**; **Mixed / augmented reality**; **Interactive systems and tools**.

KEYWORDS

Gaze Visualisation, Human-Computer Interaction, Mixed Reality Remote Collaboration, CSCW

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1 INTRODUCTION

This demonstration presents a new way to represent gaze cues in Mixed Reality (MR) remote collaboration, and so enhances the collaborative experience. Gaze cues in face-to-face communication are used to share complex social signals [3], predict another person's intention [2], understand the desire to communicate [9] or seek for approval [5]. Exchanging a range of common gaze behaviours [19] such as fixations, saccades, or blinks are typically used to achieve successful face-to-face communication.

Researchers have conducted several studies to explore gaze visualisations using traditional 2D displays for remote collaboration, but

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the limited screen space and poorly conveyed spatial cues also bring more challenges including confusion and distraction [20][15][4]. Gaze cues have been used in Mixed Reality (MR) collaboration studies (between participants in Augmented Reality (AR) and Virtual Reality (VR)) [6], but gaze is often uni-directional [13][8][17][1]. In MR, gaze cues are often illustrated as a simple marker representing the current fixation location [10][12] or as a ray converging on a surface of the target [14][1]. Therefore, there is a research gap between the need for supporting gaze behavioural visualisations via immersive interfaces and the lack of relevant studies.

To investigate the impact of gaze behavioural visualisations in immersive applications, we developed *eyemR-Vis* (Fig. 1) a system that depicts bi-directional gaze behavioural information using natural and subtle MR visualisations for pair collaboration. While supporting explicit gaze visualisations, the system aims to reduce distracting characteristics such as busy and rigid gaze visual feedback that has been used in previous research. We expect the prototype system to open up more opportunities to investigate how visualising and sharing dynamic gaze behaviours can lead to better practices in MR remote task collaboration.

2 SYSTEM DESIGN

eyemR-Vis is a 360 panoramic live video-based MR remote collaboration system that translates various gaze behavioural cues into bi-directional virtual visualisations. The local host (the user in the physical task space) shares and receives verbal and gaze communication cues using an AR headset as well as his/her surroundings using a live 360 panoramic video, to and from a remote collaborator in VR. In Fig. 1, the local host wears a see-through AR head-mounted display (HMD) with an eye-tracker and a 360 panoramic video capturing device placed at the centre of the task space, while the remote collaborator views the shared live scene using a VR HMD with an eye-tracker. Collaborators can exchange their gaze cues via eye-trackers built into their HMDs in a bi-directional manner. The system is programmed to assist pair tasks between users who are not physically co-located in the same task space by explicitly adding visualisations of gaze behaviour cues from both sides into the collaborative MR interface. Through our system, both the AR and VR users know exactly where their collaborator is looking and the type of gaze behaviour they are exhibiting.

2.1 360 Mixed Reality View

To provide the remote collaborator with an independent omnidirectional natural view that simulates the task space as close as possible to the local host's environment, we position a live 360 camera in the centre of the task space. The camera captures and shares the environment as well as the local user's actions during the collaboration (Fig. 1 top left and Fig 2) - a1). The shared panoramic video enables the remote collaborator in VR to freely determine his/her own view and probe the surroundings independently from the local host's control. Meanwhile, the system also allows the VR user to zoom-in (Fig. 2 - a2) by toggling a second camera view that is placed forward from where the centre camera faces. This offers a close-up viewpoint to check objects of interests to compensate for the remote collaborator not being able to change position. The goal

is for the VR user feel that s/he is standing in the same real environment as the local AR user while being able to have independent viewpoint control.

2.2 Gaze Cues Visualisation and Interaction

In our system, we use a combination of a virtual gaze ray [7][18][14] and a cursor [12][8] which is placed where the ray and a physical surface intersect to clearly indicate the target of interest [11]. Additionally, to make gaze representation more natural we also add a virtual avatar head from where the ray starts. In the local host view, the remote user's avatar head sits where the 360 Camera is placed (Fig. 1 - bottom left), while the remote collaborator sees it superimposed on top of the local host's HMD in the 360 live video (Fig. 1 - bottom middle). For both users, their own gaze is presented only as a cursor to reduce distraction [20] and mental load [12].

We have designed four different visualisations styles to synchronise and amplify gaze behavioural cues between collaborators: 1) Browsing state (Fig. 2 - a1 and a2): defined as active eye movements such as saccades, blink, or shift in purple (own gaze) and blue (partner's gaze); 2) Focus state (Fig. 2 - b1): when the user's gaze dwells on the same point for 500 ms or more [16]. In this state the gaze visualisation turns into a yellow colour; 3) Mutual state (Fig. 2 - b2): we change both collaborators gaze visualisations green to indicate gaze overlap (joint gaze awareness); 4) Fixated circle-map (Fig. 2 - b3) : we use a semi-transparent coloured circle (8% opacity) to map the more recent ten focus locations for each collaborator. This visualisation is a variation of the traditional heatmap design where colour coding is used to represent dwell duration. To avoid distraction we limit the colour to one and use the increase of opacity to emphasise gaze dwell duration (8% every 500ms to a maximum of 80% opacity). We also limit the numbers of circles visualised to only the most recent ten per collaborator to avoid visual clutter as it may discourage collaborators from looking at the same point together [4].

2.3 User Interviews and Design Iteration

Subsequently, we also tested it with several sample users during the process of design and development of the prototype and gleaned early feedback from interviews for design iterations. We discovered that the amount of displayed gaze information can possibly increase user distraction, and thus we improved the gaze interfaces in the following ways: 1) Eliminated the real-time trajectory gaze trail and replace it with the most recent ten fixated circle-maps; 2) Enabled the choice of a stacked circle-map (circles accumulate on top of each other) or in a single layer form (circles won't be generated if the current location has one) based on the task properties; 3) Included an option to enable/disable the user's personal gaze cursor (own gaze indicator); 4) Appended a mutual state colour indicator (as described in Section 2.2) which changes the collaborators' respective gaze colours (blue and purple) to green when they overlap (joint gaze).

3 IMPLEMENTATION

The prototype system is built with the Ricoh Theta V¹ 360 panorama camera (1080p) to capture the task space of the local environment.

¹<https://theta360.com/en/about/theta/v.html>

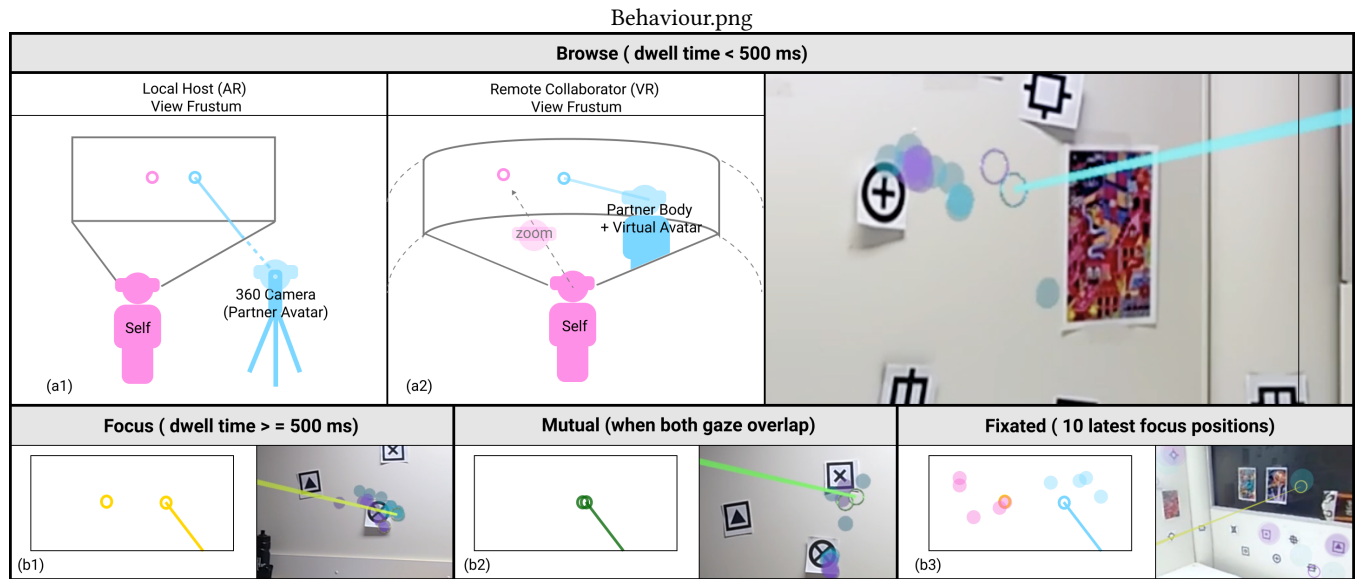


Figure 2: Visualisation of gaze cues in illustration (left) and in prototype demo (right): (a1) gaze browsing state via local host view; (a2) gaze browsing state in the remote collaborator view; (b1) gaze focus state; (b2) mutual gaze state; (b3) fixation circle-map

Both AR and VR applications are developed using the Unity game engine (v 2019.3.15) ². The local host is equipped with a Hololens2 ³ and uses a Vuforia (v9.3.3) ⁴ marker attached to the 360 camera to align both local and remote coordinate systems. The remote collaborator utilises Vive Pro Eye ⁵ running on a PC (Intel Core i7-6700 3.40GHz CPU, 16GB DDR4 RAM, NVIDIA GeForce 950 GPU) with Windows 10 OS. Tracking information is shared between AR and VR interfaces over Wifi via a standard TCP/IP connection. We used MRTK 2.4.0 ⁶ and SRanipal SDK ⁷ to create visualisations on Hololens and Vive respectively. Additionally, SteamVR Unity plugin (v2.6.1) ⁸ is used to support PC-Controller-HMD interactions such as zoom or check reference.

To achieve a seamless coordinate system alignment, we firstly manually align a virtual sphere to the 360 camera's lens relative to the Vuforia tracking marker. When the Hololens on the local host side initially tracks the Vuforia marker, it records the position and orientation of the sphere (i.e. the centre of the 360 camera) in the AR tracking coordinate frame. The system then transmits the tracking information of target objects (e.g. AR user's HMD and the gaze point) in relation to the AR coordinate system to the VR user over Wifi. On the VR side, the system mathematically recalculates the transmitted position and orientation of target objects in accordance with the VR coordinate system which maps the objects onto the spherical surface of the panorama video. We disable the HMD head position tracking by fixing it at the centre of the live panoramic

footage shared from the task space. Subsequently, we use Hololens2 and Vive Pro Eye built-in eye trackers to capture gaze data. While the gaze point tracked on the Hololens2 is shared and mapped using the method described above, the gaze point of the VR user is converted as a direction vector when shared to the AR side. The Hololens2 then uses this direction vector to cast a ray from the centre of the 360 camera onto the physical surfaces reconstructed through its spatial mapping feature, and the hit point is calculated to indicate the VR user's gaze position in the 3D task space.

4 USER EXPERIENCE

Users can play two roles a) local host and b) remote collaborator to perform two types of visual search tasks: a) symmetric - a task where both collaborators are at the same knowledge level hence work closely together to find a target, and b) asymmetric - where the remote collaborator is at a better knowledge level to give instructions to help local host solve his/her problem. In the local host role, users are able to access all communication cues such as gaze, speech, gesture, and hand pointer etc while performing tasks in his/her immediate work space. For instance, if a local host wants to deliver information to the remote collaborator, s/he can point at a specific location using either gaze pointer with verbal instruction or walk up to the target of interest to point by hand; When the local user needs to receive information from the partner, s/he can also refer to the remote collaborator's virtual avatar and gaze ray to quickly narrow down the search area of the target object based on the avatar's orientation and direction, and then identify and react to the target more effectively. In the remote collaborator role, on the other hand, users view a live 360 panoramic video streaming of both the local host and his/her surroundings while sharing gaze and speech cues to the local host. Examples include probing

²<https://unity.com/>

³<https://www.microsoft.com/en-us/hololens>

⁴<https://developer.vuforia.com>

⁵<https://www.vive.com/au/product/vive-pro-eye/overview>

⁶<https://microsoft.github.io/MixedRealityToolkit-Unity/Documentation/WelcomeToMRTK.html>

⁷<https://developer.vive.com/resources/vive-sense/sdk/vive-eye-tracking-sdk-sranipal>

⁸https://valvesoftware.github.io/steamvr_unity_plugin

the shared 360 panoramic task space with the option to zoom into an omni-directional view, communicate with the local host using voice and gaze cues while viewing the partner's actions through real-life video scenes. During the collaboration, users can see their partner's avatar superimposed on each other's head (in the AR view the 360 camera is the partner's "head") and their gaze visualised as virtual cues, leading to a more engaging collaborative experience. Further, four gaze behavioural visualisations can also assist task coordination between collaborators, such as identifying the target object effectively, reacting to the partner's gaze focus quickly, or stimulating more gaze collaboration towards the mutual target of interest, etc.

5 CONCLUSION AND FUTURE WORK

In this demonstration, we introduce eyemR-Vis, a live 360 panoramic video based Mixed Reality system that uses different visualisations of bi-directional gaze behavioural cues between a local host and a remote collaborator to enable effective collaboration. Besides streaming 360 live video of the task space, our system also visually enables collaborators to share and interact with four different gaze behavioural visualisations including "browse", "focus", "mutual-gaze", and "fixated circle-map" coupled with the partner's virtual avatar and gaze ray. Our system is designed in hoping to make the remote collaborators feel physically co-located hence making task collaborations less physically demanding. With instant gaze visual feedback, the system can help users to better understand each other's focus (gaze awareness) and joint focus (joint gaze awareness) which will lead to more active collaborations.

In the future, we plan to improve our system through better user feedback and usability testing. These improvements will explore a number of factors including the change of gaze visualisation colour or shape, the representation of 3D gaze view (e.g. gaze ray, gaze frustum, or gaze arrow), and the addition of combining gesture and speech visualisations into the existing gaze ones etc. The ultimate goal is to explore the combination of different visual cues to support gaze visualisations, resulting in achieving a better user experience in MR remote collaboration.

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